WoSM Academy

MAT231-F27 · CALCULUS II

Triple integrals

Boxes, general solids, and applications

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Last time

- $\iint_R f\,dA$: the limit of Riemann sums; for $f \geq 0$, the volume under the graph.
- Fubini: for continuous f, an iterated integral in either order.
- General regions: type I (y between two curves) and type II (x between two curves); changing the order of integration.

Today: one more variable. Nothing new in principle — the work is in describing solids in space by inequalities.

This lecture

01 Triple integrals over boxes

02 General solids

03 Applications

04 Summary and practice

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 5.4 and 5.6.

01

Triple integrals over boxes

The triple integral

Let f be defined on a box $B = [a, b] \times [c, d] \times [r, s]$. Cut B into small boxes B_{ijk} of volume $\Delta V = \Delta x\, \Delta y\, \Delta z$ and choose sample points in them:

$$\iiint_B f(x, y, z)\, dV = \lim \sum_{i,j,k} f(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})\, \Delta V,$$

if the limit exists for all choices of sample points. Continuous functions are integrable.

- $f = 1$: $\iiint_B 1\, dV$ is the volume of B.
- $f = \rho$, a density (mass per unit volume): $\iiint_B \rho\, dV$ is the mass.
- There is no "graph" to look at any more: the graph of f lives in $\mathbb{R}^4$.

Fubini's theorem in three variables

Theorem (Fubini)

If f is continuous on $B = [a, b] \times [c, d] \times [r, s]$, then

$$\iiint_B f \, dV = \int_a^b \int_c^d \int_r^s f(x, y, z) \, dz \, dy \, dx,$$

and the same holds for each of the $3! = 6$ orders of integration.

- Integrate in one variable at a time, treating the others as constants. The innermost integral is a function of the two outer variables, and so on.
- The proof is the same regrouping of Riemann sums as for double integrals.

Examples

Example (A product)

$$\iiint_{[0,1]\times[0,2]\times[0,3]} xyz^2\,dV = \int_0^1 x\,dx \int_0^2 y\,dy \int_0^3 z^2\,dz = \frac{1}{2}\cdot 2\cdot 9 = 9.$$

Example ($\iiint (x+y+z)^2\,dV$ over the unit cube)

Expand: $\iiint x^2\,dV = \frac{1}{3}$ (three such terms), $\iiint 2xy\,dV = 2\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{2}$ (three such terms). Total: $1 + \frac{3}{2} = \frac{5}{2}$.

The cube has volume 1, so $\frac{5}{2}$ is also the **average value** of $(x+y+z)^2$ on it: in general $\bar{f} = \frac{1}{\mathrm{vol}(E)}\iiint_E f\,dV$.

02

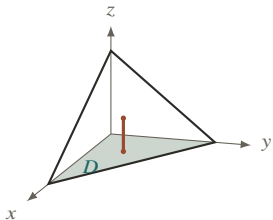
General solids

A solid between two surfaces

Let D be a region in the xy-plane, and $E = \{(x, y, z) : (x, y) \in D,\ u_1(x, y) \le z \le u_2(x, y)\}$. Then

$$\iiint_E f\, dV = \iint_D \left(\int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z)\, dz \right) dA.$$

- D is the **projection** (shadow) of E on the xy-plane.
- The double integral over D is set up as in Lecture 18. Projections on the other coordinate planes work the same way.



Over $(x, y) \in D$: $0 \le z \le 1 - x - y$.

Example: the tetrahedron

E: $x, y, z \geq 0$, $x + y + z \leq 1$. Over (x, y), z runs from 0 to $1 - x - y$; the shadow D is the triangle $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$.

$$\text{vol}(E) = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} dz\, dy\, dx = \int_0^1 \int_0^{1-x} (1 - x - y)\, dy\, dx = \frac{1}{6} \quad \text{(Lecture 18)}.$$

$$\iiint_E z\, dV = \int_0^1 \int_0^{1-x} \frac{(1-x-y)^2}{2}\, dy\, dx = \int_0^1 \frac{(1-x)^3}{6}\, dx = \frac{1}{24}.$$

So the average height of a point of E is $\frac{1/24}{1/6} = \frac{1}{4}$. By symmetry the centroid is $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$ — the average of the four vertices.

Example: a solid with a curved side

E is bounded by the parabolic cylinder $y = x^2$ and the planes $z = 0$ and $y + z = 1$.

- Over a point (x, y) of the shadow, z runs from 0 to $1 - y$.
- The shadow: $z = 0$ and $y + z = 1$ meet where $y = 1$, so D is $x^2 \le y \le 1$, $-1 \le x \le 1$.

$$\operatorname{vol}(E) = \int_{-1}^{1} \int_{x^2}^{1} \int_{0}^{1-y} dz\, dy\, dx = \int_{-1}^{1} \frac{(1-x^2)^2}{2}\, dx = \frac{1}{2}\left(2 - \frac{4}{3} + \frac{2}{5}\right) = \frac{8}{15}.$$

The same solid, in another order

Project the same solid on the yz-plane instead.

- The shadow is the triangle $0 \le y \le 1$, $0 \le z \le 1 - y$ (between $z = 0$ and $y + z = 1$).
- Over a point (y, z) of it, x runs across the parabolic cylinder: $-\sqrt{y} \le x \le \sqrt{y}$.

$$\int_0^1 \int_0^{1-y} \int_{-\sqrt{y}}^{\sqrt{y}} dx\, dz\, dy = \int_0^1 2\sqrt{y}\,(1-y)\, dy = 2\left(\frac{2}{3} - \frac{2}{5}\right) = \frac{8}{15}.$$

Same answer, as Fubini promises. Here both orders are easy; often only one is.

Slicing by planes

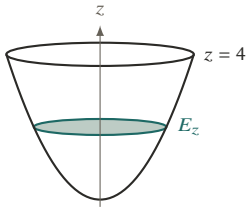
Integrate over x and y first. With E_z the slice of E at height z:

$$\iiint_E f\,dV = \int_r^s \left(\iint_{E_z} f\,dA \right) dz, \qquad \text{vol}(E) = \int_r^s \text{area}(E_z)\,dz.$$

Example (Under $z = 4$, above $z = x^2 + y^2$)

The slice at height z is the disc $x^2 + y^2 \le z$, of area πz: $\text{vol} = \int_0^4 \pi z\,dz = 8\pi$.

Projecting on the xy-plane instead leads to an integral over a disc, best done in polar coordinates (Lecture 20).



The paraboloid cut off at $z = 4$.

Choosing the order

- Sketch the solid, or at least its shadows on the coordinate planes.
- Choose as the inner variable one that runs between **two** surfaces, each given by a single formula, for every point of the shadow. Otherwise, split the solid.
- Of the six orders, prefer the one whose inner integrals are easy — as in two variables, a change of order can turn an impossible integral into an easy one.
- If every slice E_z is a simple shape (a disc, a square, a triangle), slice by planes.

> **Round solids**
>
> Balls, cylinders and cones lead to square roots in Cartesian limits. Lecture 20 introduces cylindrical and spherical coordinates for them.

03

Applications

Mass, centre of mass, moment of inertia

A solid E with density $\rho(x, y, z)$:

$$M = \iiint_E \rho \, dV, \qquad \bar{x} = \frac{1}{M} \iiint_E x\rho \, dV \ \ (\bar{y}, \bar{z} \text{ alike}), \qquad I_z = \iiint_E (x^2 + y^2)\, \rho \, dV.$$

I_z, the moment of inertia about the z-axis, measures resistance to rotation about it.

Example (The unit cube $[0,1]^3$)

- Density $\rho = z$ (heavier at the top): $M = \frac{1}{2}$, $\bar{z} = \frac{1}{M} \iiint z^2 \, dV = 2 \cdot \frac{1}{3} = \frac{2}{3}$, $\bar{x} = \bar{y} = \frac{1}{2}$.
- Density $\rho = 1$, about the z-axis (an edge): $I_z = \iiint (x^2 + y^2) \, dV = \frac{2}{3}$. About the parallel axis through the centre: $\iiint \left((x - \frac{1}{2})^2 + (y - \frac{1}{2})^2\right) dV = \frac{1}{6}$.

Probability in three dimensions

A joint density $p(x, y, z) \geq 0$ with $\iiint_{\mathbb{R}^3} p \, dV = 1$ gives $P\big((X, Y, Z) \in E\big) = \iiint_E p \, dV$.

Example (X, Y, Z independent and uniform on $[0, 1]$)

Then $p = 1$ on the unit cube, and probabilities are volumes.

- $P(X + Y + Z \leq 1) = \frac{1}{6}$: the tetrahedron.
- $P(X < Y < Z) = \int_0^1 \int_0^z \int_0^y dx \, dy \, dz = \int_0^1 \frac{z^2}{2} \, dz = \frac{1}{6}$.

Check: the $3! = 6$ orderings of X, Y, Z are equally likely, by symmetry.

More dimensions

Example (In $\mathbb{R}^n$: the simplex $\Delta_n = \{x_i \geq 0,\ x_1 + \cdots + x_n \leq 1\}$)

The slice at $x_n = t$ is Δ_{n-1} scaled by $1 - t$, of volume $(1-t)^{n-1}\,\mathrm{vol}(\Delta_{n-1})$:

$$\mathrm{vol}(\Delta_n) = \mathrm{vol}(\Delta_{n-1}) \int_0^1 (1-t)^{n-1}\,dt = \frac{\mathrm{vol}(\Delta_{n-1})}{n} = \cdots = \frac{1}{n!}.$$

High dimensions

A grid with k points per axis has k^n points: hopeless for $n = 100$. Monte Carlo methods average f at N random points; the typical error is proportional to $1/\sqrt{N}$ for every n.

04

Summary and practice

Summary

- $\iiint_E f\,dV$: the limit of Riemann sums over small boxes; volume ($f = 1$), mass ($f = \rho$).
- **Fubini**: on a box, any of the six orders of iterated integration.
- A solid between two surfaces over a shadow D: $\iiint_E f\,dV = \iint_D \int_{u_1}^{u_2} f\,dz\,dA$.
- Slicing: $\mathrm{vol}(E) = \int \mathrm{area}(E_z)\,dz$.
- Applications: mass, centre of mass, moments of inertia, probabilities as volumes. The simplex in $\mathbb{R}^n$ has volume $1/n!$.

Practice

1. Compute $\iiint_B (x + yz)\,dV$ for $B = [0,1] \times [0,1] \times [0,2]$.
2. Find the volume of the tetrahedron $x, y, z \geq 0$, $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} \leq 1$ $(a, b, c > 0)$.
3. Find the volume of the solid under $z = y$ and above the region $0 \leq x \leq 1$, $0 \leq y \leq 1 - x^2$.
4. Write $\iiint_E f\,dV$ for the tetrahedron $x, y, z \geq 0$, $x + y + z \leq 1$ in the order $dx\,dy\,dz$.
5. Find the volume of a ball of radius R by slicing.
6. Find the mass and the centre of mass of the unit cube with density $\rho = x + y + z$.
7. X, Y, Z are independent and uniform on $[0,1]$. Find $P(X + Y \leq Z)$.

Answers

1. $\iiint x\,dV + \iiint yz\,dV = 1 + 1 = 2$.
2. $\frac{abc}{6}$ (substitute $x = au$, $y = bv$, $z = cw$, or integrate directly).
3. $\int_0^1 \int_0^{1-x^2} y\,dy\,dx = \int_0^1 \frac{(1-x^2)^2}{2}\,dx = \frac{4}{15}$.
4. $\int_0^1 \int_0^{1-z} \int_0^{1-y-z} f\,dx\,dy\,dz$.
5. The slice at height z is a disc of radius $\sqrt{R^2 - z^2}$: $\int_{-R}^{R} \pi(R^2 - z^2)\,dz = \frac{4}{3}\pi R^3$.
6. $M = \frac{3}{2}$; $\iiint x\rho\,dV = \frac{1}{3} + \frac{1}{4} + \frac{1}{4} = \frac{5}{6}$, so $\bar{x} = \bar{y} = \bar{z} = \frac{5}{9}$.
7. For fixed z, $\{x + y \le z\}$ is a triangle of area $\frac{z^2}{2}$: $\int_0^1 \frac{z^2}{2}\,dz = \frac{1}{6}$.

Next time

Lecture 20 — Change of variables (Unit 20)

- The change of variables formula, and the Jacobian determinant
- Polar coordinates for discs and annuli
- Cylindrical and spherical coordinates for round solids

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 5.3, 5.5 and 5.7.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.